

Reductive cleavage of Zn–C bonds by group 14 Zintl anions: synthesis and characterisation of $[\text{E}_9\text{ZnR}]^{3-}$ ($\text{E} = \text{Ge}, \text{Sn}, \text{Pb}$; $\text{R} = \text{Mes}, {}^i\text{Pr}$)[†]

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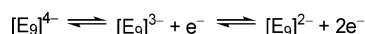
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The reaction of ethylenediamine solutions of K_4E_9 ($\text{E} = \text{Ge}, \text{Sn}, \text{Pb}$) with diaryl and dialkyl organozinc reagents ZnMes_2 and Zn^iPr_2 yielded the functionalized Zintl ions, *closo*- $[\text{E}_9\text{ZnR}]^{3-}$ ($\text{R} = \text{Mes}$: $\text{E} = \text{Ge}$ (**1**), Sn (**2**), Pb (**3**); $\text{R} = {}^i\text{Pr}$: $\text{E} = \text{Ge}$ (**4**), Sn (**5**), Pb (**6**)). These reactions proceed *via* reductive zinc–carbon bond activation of the organometallic reagents by the free solvated electrons present in ethylenediamine solutions of intermetallic Zintl phases. The functionalized cluster anions **1**, **2** and **4–6** were characterized in the solid-state as $[\text{K}(2,2,2\text{-crypt})]^+$ salts by single-crystal X-ray diffraction and elemental analysis, while the presence of all six cluster anions in solution was confirmed by ${}^1\text{H}$ and ${}^{13}\text{C}\{{}^1\text{H}\}$ NMR spectroscopy and electrospray mass spectrometry.

Introduction

Several recent studies investigating the solution reactivity of nine-atom deltahedral Zintl ions, $[\text{E}_9]^{x-}$ ($\text{E} = \text{Si}, \text{Ge}, \text{Sn}, \text{Pb}$, $x = 2\text{--}4$), have demonstrated that ethylenediamine (en) and liquid ammonia solutions of the cluster anions are capable of reductively cleaving weak M–C bonds.^{1,2} Thus, the reaction of group 14 Zintl anions with post-transition metal organometallics, such as diphenylzinc (ZnPh_2) and mesitylcopper (CuMes) have been found to yield *closo*- $[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]^{3-}$ ($\text{E} = \text{Si}, \text{Ge}, \text{Sn}, \text{Pb}$) and $[\text{Cu@E}_9]^{3-}$ ($\text{E} = \text{Sn}, \text{Pb}$), respectively.^{1,2} Such reductive metal–carbon bond activation arises as a consequence of the free solvated electrons present in solutions of the nine-atom cluster anions, which exist in equilibria in solution between the 4, 3 and 2 oxidation states (Scheme 1).³ These studies are amongst the latest in a growing number of reports contributing to the development of a better understanding of the behaviour of these highly reductive anionic clusters in solution.³



Scheme 1 Equilibrium of $[\text{E}_9]^{4-}$ ($\text{E} = \text{Si}, \text{Ge}, \text{Sn}, \text{Pb}$) clusters in ethylenediamine and liquid ammonia solutions.

The earliest solution reactivity studies on deltahedral nine atom tin and lead clusters were carried out with transition metal organometallics, such as $\text{LM}(\text{CO})_3$ ($\text{M} = \text{Cr}, \text{Mo}, \text{W}$; $\text{L} =$ mesitylene, cycloheptatriene or toluene) and demonstrated that nine atom Zintl ions are capable of replacing labile η^6 -arene ligands and coordinating to transition metal centers.^{4a} This research resulted in the isolation and characterisation of a family of anionic *closo*-clusters $[\text{E}_9\text{M}(\text{CO})_3]^{4-}$ ($\text{E} = \text{Sn}, \text{Pb}$; $\text{M} = \text{Cr}, \text{Mo}, \text{W}$).⁴ Numerous studies have since followed investigating the reactivity of $[\text{E}_9]^{4-}$ ($\text{E} = \text{Si}, \text{Ge}, \text{Sn}, \text{Pb}$) clusters towards a variety of transition metal reagents yielding interesting novel cluster geometries.^{5–15} As

a consequence, several “naked” endohedral clusters have been isolated to date including $[\text{Cu@E}_9]^{3-}$ ($\text{E} = \text{Sn}, \text{Pb}$),² $[\text{Ni@Pb}_{10}]^{2-}$,⁵ $[\text{M@Pb}_{12}]^{2-}$ ($\text{M} = \text{Ni}, \text{Pd}, \text{Pt}$),⁶ $[\text{M}_2\text{Sn}_{17}]^{4-}$ ($\text{M} = \text{Ni}, \text{Pt}$),⁷ $[\text{Ni}(\text{Ni@Ge}_9)_2]^{4-}$,⁸ and $[\text{Pd}_2\text{@E}_{18}]^{4-}$ ($\text{E} = \text{Ge}, \text{Sn}$).^{9,10} Furthermore, clusters exhibiting a combination of interstitial and functionalized surface metal atoms have also emerged as a result of this research, yielding species such as $[\text{Ni}_6\text{Ge}_{13}(\text{CO})_5]^{4-}$,¹¹ $[\text{Pt@Sn}_9\text{Pt}(\text{PPh}_3)]^{2-}$,¹² $[\text{Ni@Sn}_9\text{Ni}(\text{CO})]^{3-}$,¹² $[\text{Ni@Ge}_9\text{NiL}]^{2-}$ ($\text{L} = \text{CO}, \text{PPh}_3$),^{12,13} and $[\text{Ni@Ge}_9\text{NiL}]^{3-}$ ($\text{L} = \text{en}, -\text{C}\equiv\text{CPh}$).¹³ Most recently, reactions with post-transition metal reagents such as $\text{AuCl}(\text{PPh}_3)$ have yielded unprecedented mixed-metal cluster geometries such as those observed for $[\text{Au}_3(\text{Ge}_9)_2]^{5-}$ or $[\text{Au}_3\text{Ge}_{45}]^{9-}$.^{14,15}

Among the entire family of group 14 Zintl ions, some highly specific reactivity has been observed for the nine atom clusters of germanium. Investigations into the oxidative coupling of germanium Zintl anions $[\text{Ge}_9]^{4-}$ have yielded a series of discrete molecular species where the cluster ions are linked together *via* *exo*-bonds to form dimers,¹⁶ trimers,¹⁷ tetramers,¹⁸ and extended chains.¹⁹ Similarly, when oxidized by post-transition metal reagents, metal-linked chains such as $\{[\text{Hg}(\text{Ge}_9)]^{2-}\}_\infty$ and discrete molecular rods such as $[\text{Hg}_3(\text{Ge}_9)_4]^{10-}$ have also become available.^{20,21} Related *exo*-bonded clusters can also be isolated when employing main group nucleophilic reagents. These studies have given rise to a series of novel functionalized cluster species exhibiting *exo*-bonded main group substituents. Thus, the reaction of $[\text{Ge}_9]^{4-}$ clusters towards main group nucleophiles (Nu^-) such as $(\text{C}_6\text{H}_5)_3\text{E}^-$ ($\text{E} = \text{Sb}, \text{Bi}$) and R_3E^- ($\text{E} = \text{Ge}, \text{Sn}$; $\text{R} = \text{C}_6\text{H}_5, \text{CH}_3$) have resulted in the isolation of functionalized clusters such as $[\text{Ge}_9\text{-Nu}]^{3-}$, $[\text{Nu-Ge}_9\text{-Nu}]^{2-}$ and $[\text{Nu-Ge}_9\text{-Ge}_9\text{-Nu}]^{4-}$.²² The insights into cluster reactivity uncovered by this research have recently been employed to yield organically functionalized germanium cluster anions exhibiting direct cluster–carbon bonds such as $[\text{R-Ge}_9\text{-R}]^{2-}$ ($\text{R} = -\text{CH}=\text{CH}-\text{C}_5\text{H}_4\text{FeCp}, -\text{CH}=\text{CH}_2$),²³ $[\text{R-Ge}_9\text{-Ge}_9\text{-R}]^{4-}$ ($\text{R} = {}^i\text{Bu}, {}^n\text{Bu}, {}^s\text{Bu}, {}^t\text{Am}$),²⁴ and most recently the first examples of organically functionalized tin Zintl anions $[\text{Sn}_9\text{-R}]^{3-}$ ($\text{R} = {}^i\text{Bu}, -\text{CH}=\text{CH}_2, -\text{CH}=\text{CHPh}$).²⁵

Finally, one of the most interesting recent developments in the field of Zintl cluster chemistry is the discovery that deltahedral

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Zintl anions may be employed as precursors towards the synthesis of novel clathrates and nanostructured mesoporous materials.²⁶ These studies have brought the chemistry of Zintl ions back full circle to its origins in the solid-state, employing precursor alloys and solution methodologies to yield a series of novel extended solids.

The research described above is evidence of the polyfaceted nature of the solution reactivity of deltahedral group 14 Zintl ions. Our research with such anions is currently focused at exploring the reactivity of these nine-atom clusters towards a series of organometallic reagents at the post-transition metal/main-group divide. Herein, we report the results of recent studies employing dialkyl and diaryl organozinc reagents, which have yielded a family of functionalized clusters *closo*-[E₉ZnR]³⁻ (E = Ge, Sn, Pb; R = Mes, ⁱPr).

Experimental

General synthetic methods

All manipulations were carried out under an inert atmosphere using standard Schlenk-line and/or glovebox techniques. The intermetallic precursors, K₄E₉ (E = Ge, Sn, Pb), were synthesised according to previously reported synthetic procedures from stoichiometric mixtures of the elements (K 99.95%, Aldrich; Ge 99.999%, Strem; Sn 99.8%, Strem; Pb 99.99%, Alfa-Aesar) heated to 900 °C for 72 h in sealed niobium containers.²⁷ In order to avoid oxidation of the niobium vessels at such high temperatures, they were jacketed in flame-sealed silica ampoules under vacuum prior to heating. Ethylenediamine (99%, Aldrich) was distilled over sodium metal. Toluene (99.9%, Rathburn Chemicals Ltd.) was used after purification through an MBraun MB SPS-800 solvent system. d₅-Pyridine (99.5%, Cambridge Isotope Laboratories, Inc.) was dried over CaH₂ and vacuum distilled. All solvents were stored in gas-tight ampoules under argon. 2,2,2-crypt (4,7,13,16,21,24-hexaoxa-1,10-diazabicyclo[8.8.8]-hexacosane, Acros, 98%) was used as received after carefully drying under vacuum. Dimesitylzinc, ZnMes₂,²⁸ and diisopropylzinc, ZnⁱPr₂,²⁹ were synthesised using previously reported methods. Dimesitylzinc was dried under vacuum and stored in a glovebox, whilst diisopropylzinc was kept under an inert atmosphere at -20 °C.

[K(2,2,2-crypt)]₃[Ge₉ZnMes]. K₄Ge₉ (62 mg, 0.076 mmol), 2,2,2-crypt (103 mg, 0.275 mmol) and ZnMes₂ (35 mg, 0.114 mmol) were weighed out into a Schlenk tube in a glovebox. The Schlenk tube was subsequently transferred to a vacuum line where approximately 2 mL of ethylenediamine were added *via* a cannula. The reaction mixture was stirred for 1 h, after which the resulting dark red-orange solution was filtered and the filtrate layered with toluene to allow for crystallisation. Large dark red-orange rod-shaped crystals suitable for single-crystal X-ray diffraction were obtained after 2 d (48% crystalline yield). Anal. calcd for C₆₃H₁₁₉Ge₉K₃N₆O₁₈Zn: C 36.27, H 5.75, N 4.03. Found: C 36.18, H 5.81, N 4.09. ¹H NMR (299.86 MHz, d₅-pyridine) δ/ppm: 6.95 (s, *m*-C₉H₁₁), 3.78 (s, *o*-CH₃), 3.42 (s, 2,2,2-crypt), 3.35 (t, ³J_{H-H} = 6 Hz, 2,2,2-crypt), 2.32 (t, 2,2,2-crypt), 2.19 (s, *p*-CH₃). ¹³C{¹H} NMR (75.49 MHz, d₅-pyridine) δ/ppm: 172.3 (*i*-C₉H₁₁), 144.9 (*o*-C₉H₁₁), 132.1 (*m*-C₉H₁₁), 126.6 (*p*-C₉H₁₁), 70.9 (2,2,2-crypt), 68.1 (2,2,2-crypt), 54.3 (2,2,2-crypt), 31.0 (*o*-CH₃), 22.2 (*p*-CH₃). ES-MS (ES-): *m/z* 838.48 {[Ge₉Zn(C₉H₁₁)]⁻},

1253.78 {[K(2,2,2-crypt)][Ge₉Zn(C₉H₁₁)]⁻}, 1669.04, {[K(2,2,2-crypt)]₂[Ge₉Zn(C₉H₁₁)]⁻}. ES-MS (ES+): *m/z* 2499.95 {[K(2,2,2-crypt)]₄[Ge₉Zn(C₉H₁₁)]⁺}.

[K(2,2,2-crypt)]₃[Sn₉ZnMes].tol. K₄Sn₉ (80 mg, 0.066 mmol), 2,2,2-crypt (103 mg, 0.273 mmol) and ZnMes₂ (30 mg, 0.099 mmol) were weighed out into a dry Schlenk tube in a glovebox. The reaction vessel was transferred to a Schlenk line under an inert atmosphere and approximately 2 mL of ethylenediamine was used to dissolve the precursors. The reaction mixture was stirred for 1 h, after which the resulting dark yellow-brown solution was filtered and the filtrate layered with toluene to allow for crystallisation. Small dark brown-black brick-shaped crystals suitable for single-crystal X-ray diffraction were obtained after 2 d (23%, crystalline yield). Anal. calcd for C₇₀H₁₂₇K₃N₆O₁₈Sn₉Zn: C 32.42, H 4.94, N 3.24. Found: C 30.09, H 4.72, N 3.26. ¹H NMR (299.87 MHz, d₅-pyridine) δ/ppm: 6.99 (s, *m*-C₉H₁₁), 3.85 (s, *o*-CH₃), 3.32 (s, 2,2,2-crypt), 3.26 (broad s, 2,2,2-crypt), 2.29 (broad s, 2,2,2-crypt), 2.19 (s, *p*-CH₃). ¹³C{¹H} NMR (75.49 MHz, d₅-pyridine) δ/ppm: 142.1 (*o*-C₉H₁₁), 130.8 (*m*-C₉H₁₁), 126.2 (*p*-C₉H₁₁), 70.2 (2,2,2-crypt), 69.3 (2,2,2-crypt), 55.7 (2,2,2-crypt), 31.4 (*o*-CH₃), 21.03 (*p*-CH₃). ES-MS (ES-): *m/z* 1251.93 {[Sn₉Zn(C₉H₁₁)]⁻}, 1667.17 {[K(2,2,2-crypt)][Sn₉Zn(C₉H₁₁)]⁻}, 2083.43 {[K(2,2,2-crypt)]₂[Sn₉Zn(C₉H₁₁)]⁻}. ES-MS (ES+): *m/z* 2917.02 {[K(2,2,2-crypt)]₄[Sn₉Zn(C₉H₁₁)]⁺}.

[K(2,2,2-crypt)]₃[Pb₉ZnMes]. K₄Pb₉ (159 mg, 0.079 mmol), 2,2,2-crypt (101 mg, 0.268 mmol) and ZnMes₂ (27 mg, 0.088 mol) were weighed out into a Schlenk tube in a glovebox. The closed Schlenk tube was subsequently transferred under an inert atmosphere to a Schlenk line where approximately 2 mL of ethylenediamine was used to dissolve the reagents. The reaction mixture was stirred for 1 h, after which the resulting dark brown-black solution was filtered and the filtrate layered with toluene to allow for crystallisation (31% crystalline yield). Anal. calcd for C₆₃H₁₁₉K₃N₆O₁₈Pb₉Zn: C 22.94, H 3.64, N 2.55. Found: C 22.85, H 3.70, N 2.52. ¹H NMR (300.17 MHz, d₅-pyridine) δ/ppm: 7.28 (s, *m*-C₉H₁₁), 3.86 (s, *o*-CH₃), 3.42 (s, 2,2,2-crypt), 3.35 (t, ³J_{H-H} = 5 Hz, 2,2,2-crypt), 2.34 (t, 2,2,2-crypt), 2.18 (s, *p*-CH₃). ¹³C{¹H} NMR (75.49 MHz, d₅-pyridine) δ/ppm: 177.4 (*i*-C₉H₁₁), 142.5 (*o*-C₉H₁₁), 134.2 (*m*-C₉H₁₁), 131.4 (*p*-C₉H₁₁), 71.4 (2,2,2-crypt), 68.6 (2,2,2-crypt), 54.7 (2,2,2-crypt), 29.9 (*o*-CH₃), 23.3 (*p*-CH₃). ES-MS (ES-): *m/z* 2049.95 {[Pb₉Zn(C₉H₁₁)]⁻}, 2465.14 {[K(2,2,2-crypt)][Pb₉Zn(C₉H₁₁)]⁻} and 2882.22 {[K(2,2,2-crypt)]₂[Pb₉Zn(C₉H₁₁)]⁻}.

[K(2,2,2-crypt)]₃[Ge₉ZnⁱPr].2tol.en. K₄Ge₉ (61 mg, 0.074 mmol) and 2,2,2-crypt (101 mg, 0.267 mmol) were weighed out into a Schlenk tube in a glovebox. The Schlenk tube was then transferred to a Schlenk line and approximately 2 mL of ethylenediamine was transferred *via* a cannula. After 5 min of stirring, ZnⁱPr₂ (24 μL, 0.158 mmol) was added *via* a microsyringe. The reaction mixture was stirred for 1 h, after which the resulting dark red-orange solution was filtered and the filtrate layered with toluene to allow for crystallisation. Long clear orange-yellow crystals suitable for single-crystal X-ray diffraction were obtained after 2 d (58% crystalline yield). Anal. calcd for C₇₃H₁₃₉Ge₉K₃N₈O₁₈Zn: C 38.89, H 6.22, N 4.97. Found: C 38.83, H 6.08, N 5.04. ¹H NMR (300.17 MHz, d₅-pyridine)

δ/ppm : 3.48 (s, 2,2,2-crypt), 3.37 (t, $^3J_{\text{H-H}} = 5 \text{ Hz}$, 2,2,2-crypt), 2.35 (d, $^3J_{\text{H-H}} = 8 \text{ Hz}$, 6H, $\text{CH}(\text{CH}_3)_2$), 2.34 (t, 2,2,2-crypt), 1.36 (sep, $\text{CH}(\text{CH}_3)_2$). $^{13}\text{C}\{^1\text{H}\}$ NMR (75.49 MHz, d_5 -pyridine) δ/ppm : 71.0 (2,2,2-crypt), 68.2 (2,2,2-crypt), 54.4 (2,2,2-crypt), 36.8 ($\text{CH}(\text{CH}_3)_2$), 27.5 ($\text{CH}(\text{CH}_3)_2$). ES-MS (ES $^-$): m/z 762.20 $\{[\text{Ge}_9\text{Zn}^i\text{Pr}]^-\}$, 1177.36 $\{[\text{K}(2,2,2\text{-crypt})][\text{Ge}_9\text{Zn}^i\text{Pr}]^-\}$, 1592.53 $\{[\text{K}(2,2,2\text{-crypt})]_2[\text{Ge}_9\text{Zn}^i\text{Pr}]^-\}$. ES-MS (ES $^+$): m/z 2428.03 $\{[\text{K}(2,2,2\text{-crypt})]_4[\text{Ge}_9\text{Zn}^i\text{Pr}]^+\}$.

[K(2,2,2-crypt)]₃[Sn₉ZnⁱPr]·0.5en. K₄Sn₉ (84 mg, 0.069 mmol) and 2,2,2-crypt (106 mg, 0.2804 mmol) were weighed out into a Schlenk tube inside a glovebox. The Schlenk was then transferred under an inert atmosphere to a Schlenk line. Approximately 2 mL of ethylenediamine was transferred to the Schlenk using a cannula giving rise to a dark red-brown solution. After stirring for 5 min, ZnⁱPr₂ (19 μL , 0.125 mmol) was added using a microsyringe. The reaction mixture was stirred for 1 h, after which the resulting dark brown-yellow solution was filtered and the filtrate layered with toluene to allow for crystallisation. Large dark orange-yellow plate-like crystals suitable for single-crystal X-ray diffraction were obtained after 2 d (27% crystalline yield). Anal. calcd for C₃₈H₁₁₉K₃N₇O₁₈Sn₉Zn: C 28.37, H 4.89, N 4.00. Found: C 28.34, H 4.90, N 3.95. ^1H NMR (300.17 MHz, d_5 -pyridine) δ/ppm : 3.43 (s, 2,2,2-crypt), 3.36 (t, $^3J_{\text{H-H}} = 5 \text{ Hz}$, 2,2,2-crypt), 2.51 (d, $^3J_{\text{H-H}} = 8 \text{ Hz}$, $\text{CH}(\text{CH}_3)_2$), 2.35 (t, 2,2,2-crypt), 1.09 (sep, $\text{CH}(\text{CH}_3)_2$). $^{13}\text{C}\{^1\text{H}\}$ NMR (75.49 MHz, d_5 -pyridine) δ/ppm : 71.2 (2,2,2-crypt), 68.3 (2,2,2-crypt), 60.1 ($\text{CH}(\text{CH}_3)_2$), 54.5 (2,2,2-crypt), 25.3 ($\text{CH}(\text{CH}_3)_2$). ES-MS (ES $^-$): m/z 1175.68 $\{[\text{Sn}_9\text{Zn}^i\text{Pr}]^-\}$, 1590.78 $\{[\text{K}(2,2,2\text{-crypt})][\text{Sn}_9\text{Zn}^i\text{Pr}]^-\}$, 2006.78 $\{[\text{K}(2,2,2\text{-crypt})]_2[\text{Sn}_9\text{Zn}^i\text{Pr}]^-\}$. ES-MS (ES $^+$): m/z 2838.99 $\{[\text{K}(2,2,2\text{-crypt})]_4[\text{Sn}_9\text{Zn}^i\text{Pr}]^+\}$.

[K(2,2,2-crypt)]₆[Pb₉ZnⁱPr]₂·en. K₄Pb₉ (158 mg, 0.079 mmol) and 2,2,2-crypt (107 mg, 0.284 mmol) were weighed out into a Schlenk tube in a glovebox. The reaction vessel was transferred to a Schlenk line and the reagents dissolved in approximately 2 mL of ethylenediamine. After stirring for approximately 5 min, ZnⁱPr₂ (20 μL , 0.132 mmol) was added *via* a microsyringe. The reaction mixture was stirred for 2–3 h, after which the resulting dark black-brown solution was filtered and the filtrate layered with toluene to allow for crystallisation. Large, black plate-like crystals suitable for single-crystal X-ray diffraction were obtained after 2 d (27% crystalline yield). Anal. calcd for C₁₁₆H₂₃₈K₆N₁₄O₃₆Pb₁₈Zn₂: C 21.42, H 3.69, N 3.02. Found: C 21.51, H 3.59, N 2.92. ^1H NMR (300.17 MHz, d_5 -pyridine) δ/ppm : 3.42 (s, 2,2,2-crypt), 3.35 (t, $^3J_{\text{H-H}} = 5 \text{ Hz}$, 2,2,2-crypt), 2.83 (d, $^3J_{\text{H-H}} = 7 \text{ Hz}$, $\text{CH}(\text{CH}_3)_2$), 2.35 (t, 2,2,2-crypt), 0.03 (sep, $\text{CH}(\text{CH}_3)_2$). $^{13}\text{C}\{^1\text{H}\}$ NMR (75.49 MHz, d_5 -pyridine) δ/ppm : 71.1 (2,2,2-crypt), 70.4 ($\text{CH}(\text{CH}_3)_2$), 68.6 (2,2,2-crypt), 54.7 (2,2,2-crypt), 17.5 ($\text{CH}(\text{CH}_3)_2$). ES-MS (ES $^-$): m/z 1974.17 $\{[\text{Pb}_9\text{Zn}^i\text{Pr}]^-\}$, 2389.14 $\{[\text{K}(2,2,2\text{-crypt})][\text{Pb}_9\text{Zn}^i\text{Pr}]^-\}$.

Structure determination†

Single-crystal X-ray diffraction data were collected on an Enraf-Nonius Kappa CCD diffractometer equipped an Oxford Cryosystems low-temperature device. The crystals were selected under Paratone-N oil, mounted on micromount loops and positioned in the cold stream of the diffractometer. Data were collected at 150 K using graphite monochromatic MoK α radiation; equivalent

Table 1 Selected X-ray data collection and refinement parameters for [K(2,2,2-crypt)]₃**1** and [K(2,2,2-crypt)]₃**2**·tol

Compound	[K(2,2,2-crypt)] ₃ 1	[K(2,2,2-crypt)] ₃ 2 ·tol
FW/g mol ⁻¹	2084.62	2590.84
Space group, <i>Z</i>	<i>P</i> 2 ₁ / <i>c</i> , 4	<i>P</i> 2 ₁ / <i>c</i> , 4
<i>a</i> /Å	17.3897(1)	14.7470(1)
<i>b</i> /Å	22.4685(2)	24.6645(2)
<i>c</i> /Å	22.0943(2)	27.0957(2)
$\beta/^\circ$	98.0425(3)	101.4626(3)
<i>V</i> /Å ³	8547.79(12)	9658.86(12)
ρ_{calcd} /g cm ⁻³	1.620	1.782
Radiation, λ /Å, <i>T</i> /K	MoK α , 0.71073, 150	
μ /cm ⁻¹	3.600	2.716
Reflections collected	29 330	31 887
Independent reflections	14 916	16 791
<i>R</i> _{int}	0.0285	0.0220
<i>R</i> ₁ / <i>wR</i> ₂ , ^a <i>I</i> ≥ 2 σ _{<i>i</i>} (%)	5.61/14.21	7.62/17.39
<i>R</i> ₁ / <i>wR</i> ₂ , ^a all data (%)	7.58/15.10	9.64/18.35

^a $R_1 = [\sum \|F_o\| - |F_c|]/\sum \|F_o\|$; $wR_2 = \{[\sum w(F_o^2 - F_c^2)^2]/[\sum w(F_o^2)^2]\}^{1/2}$; $w = [\sigma^2(F_o^2) + (AP)^2 + BP]^{-1}$, where $P = [(F_o^2) + 2(F_c^2)]/3$ and the *A* and *B* values are 0.0698 and 34.71 for [K(2,2,2-crypt)]₃**1** and 0.0579 and 97.35 for [K(2,2,2-crypt)]₃**2**·tol.

Table 2 Selected X-ray data collection and refinement parameters for [K(2,2,2-crypt)]₃**4**·2tol·en, [K(2,2,2-crypt)]₃**5**·0.5en and [K(2,2,2-crypt)]₆**6**·en

Compound	[K(2,2,2-crypt)] ₃ 4 ·2tol·en	[K(2,2,2-crypt)] ₃ 5 ·0.5en	[K(2,2,2-crypt)] ₆ 6 ·en
FW/g mol ⁻¹	2252.90	2453.48	6498.28
Space group, <i>Z</i>	<i>P</i> 1̄, 2	<i>P</i> 1̄, 2	<i>P</i> -1, 2
<i>a</i> /Å	13.8650(1)	14.0006(1)	14.0881(1)
<i>b</i> /Å	15.1706(1)	14.5855(1)	25.5722(2)
<i>c</i> /Å	26.0077(2)	21.8839(2)	27.1348(2)
$\alpha/^\circ$	75.4570(10)	86.8242(4)	112.1484(3)
$\beta/^\circ$	85.3940(10)	86.3706(5)	100.0104(3)
$\gamma/^\circ$	69.1790(10)	79.6696(4)	92.2170(3)
<i>V</i> /Å ³	4949.15(6)	4383.15(6)	8858.10(12)
ρ_{calcd} /g cm ⁻³	1.512	1.859	2.436
Radiation, λ /Å, <i>T</i> /K	MoK α , 0.71073, 150		
μ /cm ⁻¹	3.116	2.987	17.501
Reflections collected	43 107	36 870	54 321
Independent reflections	22 374	19 739	30 777
<i>R</i> _{int}	0.0261	0.0294	0.0536
<i>R</i> ₁ / <i>wR</i> ₂ , ^a <i>I</i> ≥ 2 σ _{<i>i</i>} (%)	3.55/7.91	5.47/13.95	4.86/11.99
<i>R</i> ₁ / <i>wR</i> ₂ , ^a all data (%)	4.83/8.38	7.32/14.67	7.08/12.87

^a $R_1 = [\sum \|F_o\| - |F_c|]/\sum \|F_o\|$; $wR_2 = \{[\sum w(F_o^2 - F_c^2)^2]/[\sum w(F_o^2)^2]\}^{1/2}$; $w = [\sigma^2(F_o^2) + (AP)^2 + BP]^{-1}$, where $P = [(F_o^2) + 2(F_c^2)]/3$ and the *A* and *B* values are 0.0280 and 7.93 for [K(2,2,2-crypt)]₃**4**·2tol·en, 0.0273 and 75.48 for [K(2,2,2-crypt)]₃**5**·0.5en and 0.0779 and 0.00 for [K(2,2,2-crypt)]₆**6**·en.

reflections were merged, and the images were processed with the DENZO and SCALEPACK programs.³⁰ Corrections for Lorentz-polarization effects and absorption were performed, and the structures were solved by direct methods and refined on *F*² using the SHELX-97 package.³¹ Details of selected data collections and refinement parameters are given in Tables 1 and 2.

Electrospray mass-spectroscopy

Positive and negative ion mode electrospray mass spectra were recorded from DMF solutions (10–20 μM) on a Micromass LCT Time of Flight mass spectrometer with a Z-spray source (150 $^{\circ}\text{C}$ source temperature, 200 $^{\circ}\text{C}$ desolvation temperature, 2.2 kV capillary voltage, and 20 V cone voltage).

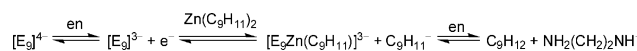
NMR

$^{13}\text{C}\{^1\text{H}\}$ and ^1H NMR spectra were recorded at room temperature in d_5 -pyridine using a Varian Mercury VX300 spectrometer (^1H 300 MHz). $^{13}\text{C}\{^1\text{H}\}$ spectra were referenced to d_5 -pyridine (δ 123.87 ppm) and ^1H NMR were referenced to the most upfield of residual protic solvent resonances (pyridine, δ 7.22 ppm).

Results and discussion

$[\text{E}_9\text{Zn}(\text{C}_9\text{H}_{11})]^{3-}$ (E = Ge (1); Sn (2); Pb(3))

Reaction of ethylenediamine solutions of K_4E_9 (E = Ge, Sn, Pb) and 2,2,2-crypt (4,7,13,16,21,24-hexaoxa-1,10-diazabicyclo[8.8.8]hexacosane) with dimethylzinc (ZnMes_2) in a 1 : 1 stoichiometric ratio (K_4E_9 : ZnMes_2) yielded the novel ZnMes functionalized clusters, *closo*- $[\text{E}_9\text{ZnMes}]^{3-}$ (E = Ge (1); Sn (2); Pb (3)). These reactions are analogous to those observed for group 14 Zintl ions with $\text{Zn}(\text{C}_6\text{H}_5)_2$,¹ and proceed *via* the reduction of one of the M–C bonds of the post-transition metal organometallic reagent by the free solvated electrons present in ethylenediamine solutions of $[\text{E}_9]^{4-}$. This results in the generation of a ZnMes moiety and the $\text{C}_9\text{H}_{11}^-$ anion, the latter of which quickly abstracts a proton from the ethylenediamine solvent yielding mesitylene and an amide, as pictured in Scheme 2. *In situ* monitoring of the crude reaction mixtures by ^1H NMR spectroscopy reveals the immediate generation of two singlet resonances corresponding to free mesitylene in addition to three resonances arising from the bonded ZnMes fragments. The functionalized cluster anions **1** and



Scheme 2 Synthesis of *closo*- $[\text{E}_9\text{ZnMes}]^{3-}$ (E = Ge, Sn, Pb).

2 (Fig. 1) were isolated as $[\text{K}(2,2,2\text{-crypt})]^+$ crystals by carefully layering the filtered ethylenediamine solutions with toluene.

Clusters **1** and **2** are best described as ten vertex deltahedral Zintl ions with bicapped square-antiprismatic geometries where one of the capping positions is occupied by a zinc metal centre. This bonding mode is entirely analogous to that observed for the aforementioned *closo*- $[\text{E}_9\text{M}(\text{CO})_3]^{4-}$ (E = Sn, Pb; M = Cr, Mo, W) and *closo*- $[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]^{3-}$ (E = Si, Ge, Sn, Pb) clusters and can be rationalized employing a delocalized bonding model.^{1,4} An analysis of these species employing Wade/Mingos electron counting rules shows that the clusters have the correct electron count to be considered *closo*-deltahedra.³² Each group 14 element contributes three orbitals and two electrons to cluster bonding and the ZnMes fragment (isolobal with BH^+) contributes three orbitals and one electron, making the total electron count for such a species $(9 \times 2) + 1 + 3 = 22$ cluster bonding electrons. This is in accordance with what would be expected for a *closo*-ten vertex cluster ($n + 1$ cluster bonding electron pairs, 22 electrons).

Bonding between $[\eta^4\text{-E}_9]^{4-}$ clusters and transition metal centers has been previously discussed in the literature for *closo*- $[\text{E}_9\text{ZnPh}]^{3-}$.¹ Theoretical calculations have shown that the clusters behave in an isolobal manner to the cyclopentadienyl anion and that they can act as formal six electron donors to metals when coordinated in η^4 -mode. A frontier orbital analysis reveals three fully occupied orbitals corresponding to the D_{4d} configuration of $[\text{E}_9]^{4-}$, which possess the correct symmetry to interact *via* a σ and two π interactions with three empty orbitals on the zinc metal centre. From a coordination chemistry viewpoint, this signifies that the electron count at the zinc centre is 18 electrons, 6 from the $[\text{E}_9]^{4-}$ cluster, 11 from the Zn^+ metal centre and one from the mesityl substituent. This bonding analysis makes the reported *closo*- $[\text{E}_9\text{ZnMes}]^{3-}$ clusters isoelectronic with species such as η^1 , η^5 -bis(pentamethylcyclopentadienyl)zinc or η^5 -cyclopentadienylzinc methyl.³³

As mentioned previously, both clusters **1** and **2** adopt bicapped square antiprismatic geometries for the $[\text{E}_9\text{Zn}]$ cluster cores, with the zinc atoms occupying capping vertices. The single-crystal X-ray data recorded for $[\text{K}(2,2,2\text{-crypt})]_3\text{2-tol}$ are of unsuitable quality for the discussion of structural metric values due to the crystallographic disorder present in the asymmetric unit at the cluster site. However, despite moderate disorder, the composition of the crystalline phase is clear. Furthermore, interatomic distance values are in roughly good agreement with those observed for the phenyl-functionalized analogue and with those observed for $[\text{Sn}_9\text{Zn}^{\text{Pr}}]^{3-}$, which will be discussed in detail in the following section.

E–E distances vary between 2.5327(10)–2.8852(11) Å for **1** (a complete list of interatomic distances is listed in Table 3). As would be expected, these distances are closely related to those observed for the analogous ZnPh functionalized ion *closo*- $[\text{Ge}_9\text{ZnPh}]^{3-}$, which vary between 2.5427(7)–2.8377(8) Å.¹ The E–Zn distances between the atoms comprising the square face of the E_9 mono-capped square antiprism and the capping zinc atom for cluster **1** vary within the range of 2.5644(11)–2.6230(12) Å. Once again these values are very similar to those corresponding to

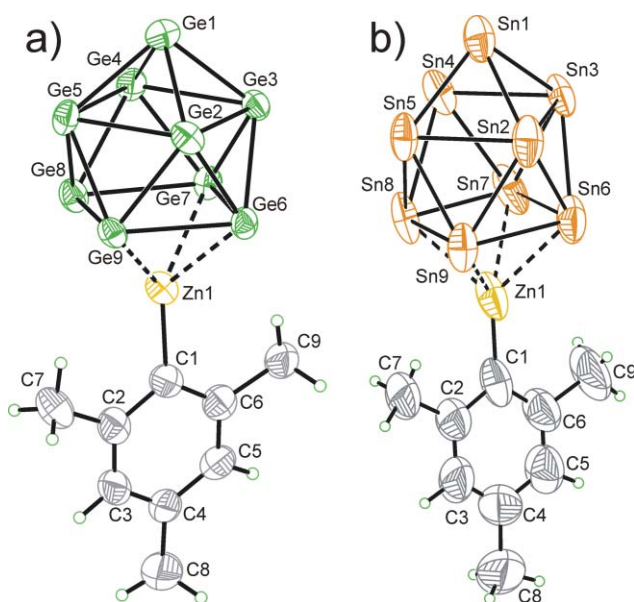


Fig. 1 Thermal ellipsoid plots of (a) **1** and (b) **2**. Anisotropic displacement ellipsoids pictured at 50% probability level.

Table 3 Selected bond lengths for clusters **1**, **4**, **5** and the two crystallographically unique clusters in $[\text{K}(2,2,2\text{-crypt})]_6[\text{Pb}_9\text{Zn}(\text{Pr})_2\text{-en}]$, **6a** and **6b**

Bond ^a	1	4	5	6a	6b
E1–E2	2.5529(11)	2.5658(4)	2.9295(8)	3.0657(5)	3.0611(6)
E1–E3	2.5486(12)	2.5665(4)	2.9515(9)	3.0835(5)	3.0836(6)
E1–E4	2.5984(12)	2.5859(4)	2.9581(8)	3.0713(6)	3.0707(5)
E1–E5	2.5801(12)	2.5712(4)	2.9405(8)	3.0514(5)	3.0638(5)
E2–E3	2.7889(11)	2.8638(4)	3.2641(8)	3.3489(5)	3.3209(5)
E2–E5	2.8066(11)	2.8428(4)	3.1787(9)	3.3237(6)	3.2997(6)
E2–E6	2.5579(12)	2.5615(4)	2.9420(9)	3.0706(5)	3.0397(6)
E2–E9	2.5703(11)	2.5570(4)	2.8955(8)	3.0718(5)	3.0691(5)
E3–E4	2.8852(11)	2.8311(4)	3.1688(9)	3.3111(5)	3.2962(6)
E3–E6	2.5431(11)	2.5572(4)	2.9561(8)	3.0833(5)	3.1079(6)
E3–E7	2.5327(10)	2.5528(4)	2.9490(8)	3.0983(6)	3.1054(5)
E4–E5	2.8181(11)	2.8245(4)	3.2167(8)	3.4250(5)	3.3970(5)
E4–E7	2.5495(11)	2.5520(4)	2.9603(9)	3.0671(5)	3.0838(6)
E4–E8	2.5613(12)	2.5519(4)	2.9543(8)	3.1059(5)	3.0984(5)
E5–E8	2.5780(11)	2.5612(4)	2.9442(9)	3.0730(6)	3.1066(5)
E5–E9	2.5659(10)	2.5626(4)	2.9316(9)	3.0244(5)	3.0632(5)
E6–E7	2.8015(10)	2.7868(4)	3.1094(8)	3.2977(5)	3.2570(5)
E6–E9	2.7687(11)	2.7674(4)	3.1321(8)	3.3179(5)	3.2852(5)
E7–E8	2.8264(11)	2.7782(4)	3.1471(8)	3.2501(5)	3.2881(5)
E8–E9	2.7693(10)	2.7494(4)	3.1739(8)	3.2325(5)	3.3049(5)
Zn1–E6	2.5644(11)	2.5999(5)	2.8066(13)	2.8445(13)	2.8513(13)
Zn1–E7	2.5930(12)	2.5915(5)	2.7702(12)	2.8013(12)	2.8332(12)
Zn1–E8	2.6177(11)	2.6225(5)	2.8095(13)	2.8574(12)	2.8470(13)
Zn1–E9	2.6230(12)	2.6132(5)	2.7776(13)	2.8471(13)	2.8347(13)
Zn1–C1	2.001(7)	2.023(3)	2.014(11)	2.021(10)	2.044(9)

^a Element numbering scheme as employed in Fig. 1 and 3.

the phenyl functionalized analogue which vary between 2.5720(8)–2.5944(9) Å for $[\text{Ge}_9\text{ZnPh}]^{3-}$. The Zn–C bond distance between the cluster and the mesityl functionality is 2.001(7) Å for cluster **1**. This value compares nicely with the Zn–C bond distances reported in the literature for molecular species exhibiting bonds between Zn and mesityl functionalities which vary between 1.942(2) and 2.041(7) Å.³⁴

Characterization of clusters **1–3** in solution was carried out by means of ^1H and $^{13}\text{C}\{^1\text{H}\}$ NMR spectroscopy. Analytically pure solid samples were dissolved in d_5 -pyridine and analyzed by ^1H NMR spectroscopy revealing three singlet resonances corresponding to the three different protic environments of a η^1 -bonded mesityl ring. These resonances could be observed at 6.95, 3.78 and 2.19 ppm for cluster **1** and at 6.99, 3.85, 2.19 and 7.28, 3.86, 2.18 ppm for clusters **2** and **3**, respectively. The $^{13}\text{C}\{^1\text{H}\}$ NMR spectra of the functionalized clusters **1** and **3** displayed four resonances in the aromatic region, corresponding to the *ipso*-, *ortho*-, *meta*- and *para*-aromatic carbon environments and the two distinct CH_3 groups of the η^1 -bonded mesityl ring. Attempts to observe the *ipso*-carbon resonance for cluster **2** proved unsuccessful due to inherent weakness of such resonances and limitations on sample concentration, however the remaining five resonances were clearly observed. Attempts to record $^{117/119}\text{Sn}$ NMR spectra for cluster **2** and ^{207}Pb NMR spectra for cluster **3** were also unsuccessful. These experiments were carried out under various temperature regimes yet no resonances were observed for either of the cluster species. We believe that this is largely due to the relatively low maximum sample concentration of the $[\text{K}(2,2,2\text{-crypt})]^+$ salts of cluster anions in d_5 -pyridine and ethylenediamine. As a result we were unable to probe whether such cluster anions exhibit fluxional behaviour in solution. Similar

functionalised clusters have been previously reported to exist in dynamic equilibrium between the η^4 and η^5 coordinated isomers, however only spectroscopic evidence for the η^4 -isomer has been observed to date.^{4b,d}

Further confirmation of the presence of the clusters in solution was obtained by negative ion mode electrospray mass-spectrometry of DMF solutions clusters **1–3**. All peaks appear as distinct mass envelopes due to the wide variety of Zn, Ge, Sn and Pb isotopes, which allows for the unequivocal assignment of cluster-based signals. As is often the case with the negative ion-mode electrospray mass-spectra of anionic Zintl ions, the clusters are observed with reduced charges as a result of the oxidation of the parent clusters during the ionization process. Furthermore, there is also evidence of extensive ion pairing between anions and the cryptated potassium cations, $[\text{K}(2,2,2\text{-crypt})]^+$. As a result, the negative ion mode spectra of the $[\text{K}(2,2,2\text{-crypt})]^+$ salts of **1–3** reveal mass envelopes corresponding to $[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]^-$ (Fig. 2; $m/z = 838.5$, 1251.9 and 2050.0 for **1**, **2** and **3**, respectively), peaks corresponding to the ion-paired species $\{[\text{K}(2,2,2\text{-crypt})][\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^-$ ($m/z = 1253.8$, 1667.2 and 2465.1 for **1**, **2** and **3**, respectively) and peaks corresponding to $\{[\text{K}(2,2,2\text{-crypt})]_2[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^-$ ($m/z = 1669.0$, 2083.5 and 2882.2 for **1**, **2** and **3**, respectively). The positive ion mode spectra of clusters **1** and **2** also reveal peaks corresponding to the $\{[\text{K}(2,2,2\text{-crypt})]_4[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^+$ ion pairs at $m/z = 2500.0$ and 2917.0, respectively. However, no such species was observed in the spectrum of the heavier lead analogue.

$[\text{E}_9\text{Zn}^{\text{Pr}}]^{3-}$ (E = Ge (**4**); Sn (**5**); Pb(**6**))

The high crystalline yields and relative synthetic ease associated with the isolation of the aforementioned Zn-aryl functionalized Zintl ions prompted us to explore whether the M–C bond activation reactions employed in their synthesis could be extended to other organometallic species. Specifically, we chose to target late transition metal organometallics bound to alkyl groups in order to determine whether such reactions could be expanded to other organometallic reagents with non-aromatic substituents. As a result, we found that reaction of ethylenediamine solutions of K_4E_9 (E = Ge, Sn, Pb) and 2,2,2-crypt (4,7,13,16,21,24-hexaoxa-1,10-diazabicyclo[8.8.8]hexacosane) with diisopropylzinc (Zn^{Pr}_2) yielded the novel Zn/Pr functionalized clusters, *closo*- $[\text{E}_9\text{Zn}^{\text{Pr}}]^{3-}$ (E = Ge (**4**); Sn (**5**); Pb (**6**); Fig. 3). Once again, reductive cleavage of one of the metal–carbon bonds in Zn^{Pr}_2 results in the generation of a Zn^{Pr} moiety, which interacts with the nine-atom clusters to yield a family of Zn-alkyl functionalized Zintl ions. To our knowledge, these reactions are the first report of the use of dialkyl organometallic reagents to generate functionalized deltahedral Zintl clusters.

X-Ray quality crystals of the $[\text{K}(2,2,2\text{-crypt})]^+$ salts of anions **4–6** (Fig. 3) can be grown by layering the filtrates of the crude reaction mixtures with toluene. Clusters **4–6** were characterised by single-crystal X-ray diffraction in $[\text{K}(2,2,2\text{-crypt})]_3[\text{Ge}_9\text{Zn}^{\text{Pr}}] \cdot 2\text{tol-en}$, $[\text{K}(2,2,2\text{-crypt})]_3[\text{Sn}_9\text{Zn}^{\text{Pr}}] \cdot 0.5\text{en}$ and $[\text{K}(2,2,2\text{-crypt})]_6[\text{Pb}_9\text{Zn}^{\text{Pr}}] \cdot \text{en}$, respectively. Once again, the structure of the three cluster anion cores can be described as bi-capped square antiprismatic with the zinc metal centers occupying capping vertices. The zinc metal completes its coordination sphere by binding to an $^{\text{Pr}}$ moiety in a two-centre, two electron bond.

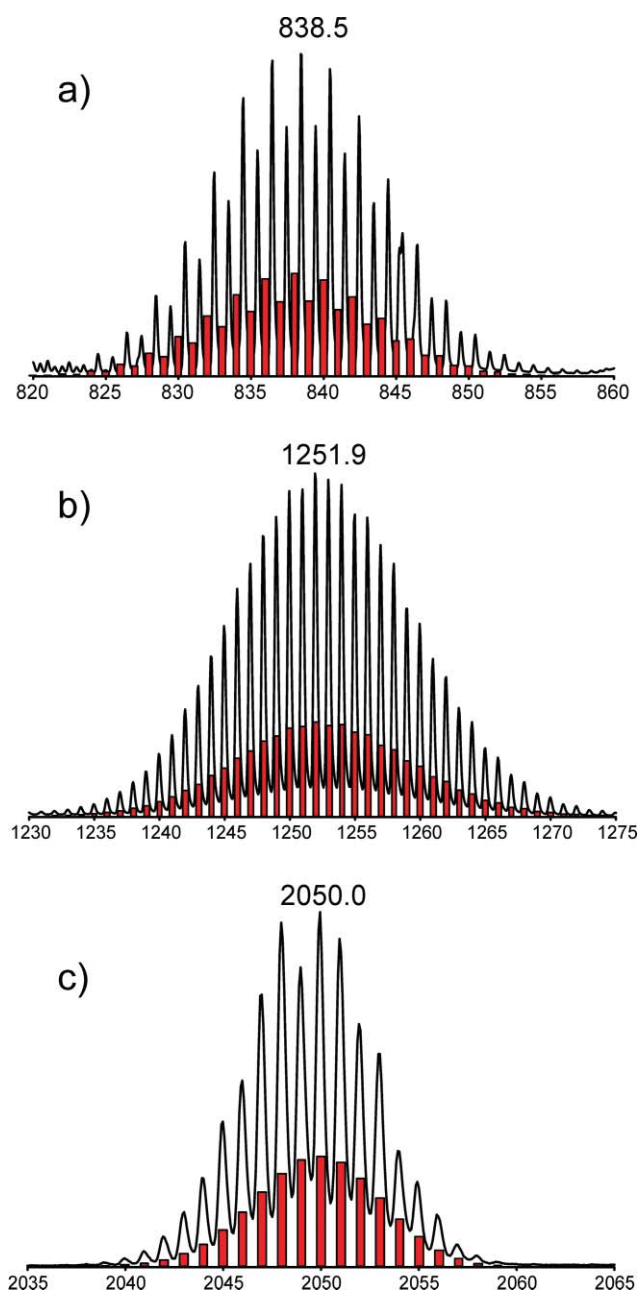


Fig. 2 Selected negative ion-mode electrospray mass-spectrometry peaks from DMF solutions containing **1**–**3**. Mass envelopes correspond to: (a) $[\text{Ge}_9\text{Zn}(\text{C}_9\text{H}_{11})]^-$, (b) $[\text{Sn}_9\text{Zn}(\text{C}_9\text{H}_{11})]^-$ and (c) $[\text{Pb}_9\text{Zn}(\text{C}_9\text{H}_{11})]^-$. Recorded experimental data are given in black with calculated isotopic distributions in red.

The E–E distances of the deltahedral cluster anions vary between 2.5519(4)–2.8638(4) Å for **4**, 2.8955(8)–3.2641(8) Å for **5** and 3.0244(5)–3.4250(5) Å and 3.0397(6)–3.3970(5) Å for the two crystallographically unique cluster anions of **6** (**6a** and **6b**) present in the asymmetric unit of $[\text{K}(2,2,2\text{-crypt})]_6[\text{Pb}_9\text{Zn}^{\text{I}}\text{Pr}]_2\cdot\text{en}$. A complete list of bond metric data is listed in Table 3 following the numbering scheme employed in Fig. 3. The interatomic distances found for cluster **4** is in good agreement with the ZnMes functionalised analogue discussed earlier (2.5427(7)–2.8377(8) Å for **1**). Likewise, all the E–E distances compare well with the values recorded for $[\text{E}_9\text{ZnPh}]^{3-}$ (E = Ge, Sn, Pb), which were found

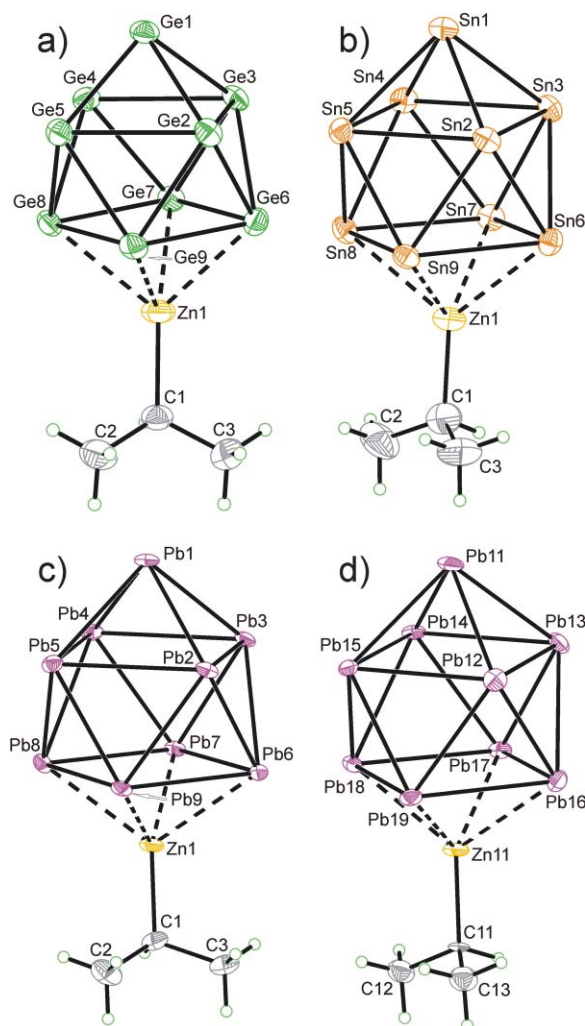


Fig. 3 Thermal ellipsoid plots of (a) **4**, (b) **5**, (c) **6a** and (d) **6b**. Anisotropic displacement ellipsoids pictured at 50% probability level.

to vary between 2.5427(7)–2.8377(8) Å, 2.9156(3)–3.2275(3) Å, and 3.0447(6)–3.4078(5) Å for the germanium, tin and lead cluster anions, respectively.¹ The distances observed between the capping Zn atoms and the E atoms comprising the square face of the monocapped square antiprism vary narrowly within the ranges 2.5915(5)–2.6225(5) Å for **4**, 2.7702(12)–2.8095(13) Å for **5** and 2.8013(12)–2.8574(12) and 2.8332(12)–2.8513(13) Å for **6a** and **6b**, respectively. Once again these values are very close to those corresponding to the phenyl functionalised analogues which vary between 2.5720(8)–2.5944(9) Å, 2.7397(4)–2.7867(4) Å and 2.7883(13)–2.8411(12) Å for $[\text{Ge}_9\text{ZnPh}]^{3-}$, $[\text{Sn}_9\text{ZnPh}]^{3-}$ and $[\text{Pb}_9\text{ZnPh}]^{3-}$, respectively.

The Zn–C bond distances between the cluster and the isopropyl functionality are 2.023(3), 2.014(11), 2.021(10) and 2.044(9) Å for clusters **4**, **5** and **6a** and **6b**, respectively. These values are on the long end of the scale of what has previously been reported for other species exhibiting Zn–isopropyl bonds, for which Zn–C distances vary between 1.978(9) and 2.032(1) Å.³⁵

Clusters **4**–**6** were characterized in solution by ^1H and $^{13}\text{C}\{^1\text{H}\}$ NMR spectroscopy. Analytically pure crystalline samples of the $[\text{K}(2,2,2\text{-crypt})]^+$ salts of the cluster anions were dissolved in d_5 -pyridine and analyzed by ^1H NMR spectroscopy revealing

two multiplet resonances corresponding to the two distinct protic environments of the isopropyl functionality. These resonances appear as a doublet for the six equivalent protons of the methyl groups and as a septet for the methine proton. The chemical shifts of the resonances were recorded at 2.35 and 1.36 ppm, 2.51 and 1.09 ppm and 2.83 and 0.03 ppm for clusters **4**, **5** and **6**, respectively (Fig. 4). The doublet resonance arising from the methyl group protons of cluster **4** was found to coincide with one of the resonances due to the 2,2,2-crypt protons, however the resolution of the NMR spectrum was sufficient to allow for the correct assignment of peak positions. The $^3J_{\text{H-H}}$ coupling constants are very similar for all three cluster anions and have values of 8 Hz for clusters **4** and **5** and 7 Hz for cluster **6**. The $^{13}\text{C}\{^1\text{H}\}$ NMR spectra of the cluster anions provide further evidence of the stability of the functionalized cluster anions in solution and exhibit two distinct singlet resonances for each of the cluster anions. These resonances occur at 36.8 and 27.5 ppm, 60.1 and 25.3 ppm, and 70.4 and 17.5 ppm for clusters **4**, **5** and **6**, respectively. The most deshielded of the resonances in each case correspond to the carbon nuclei of the methyl groups while the resonances at highest ppm values correspond to the tertiary carbons. It is interesting to note that as one descends group 14, there is an alternating effect in the shielding experienced by adjacent nuclei of the isopropyl functionalities of the $[\text{E}_9\text{Zn}^i\text{Pr}]^{3-}$ clusters. That is, as you move from germanium to lead, the first atom bonded to zinc (the tertiary carbon) experiences a deshielding effect shifting from 36.8 ppm for **4** to 60.1 ppm for **5** and finally to 70.4 ppm for **6**. The inverse effect occurs at the nuclei directly bonded to this carbon, thus the proton of the methine group experiences a shielding effect shifting from 1.36, 1.09 to 0.03 ppm for germanium, tin and lead, respectively. Likewise, the same shielding effect is observed for the primary carbon nuclei of the methyl substituents shifting from 27.5, 25.3 and 17.5 ppm for clusters **4**, **5** and **6**, respectively. Finally, the deshielding effect occurs yet again for the hydrogen nuclei of the methyl groups which shift from 2.35 to 2.51 to 2.83 ppm on moving from Ge to Sn to Pb. As with the mesityl-functionalised cluster anions **2** and **3** attempts to record the $^{117/119}\text{Sn}$ and ^{207}Pb NMR spectra for clusters **5** and **6**, respectively, proved unsuccessful.

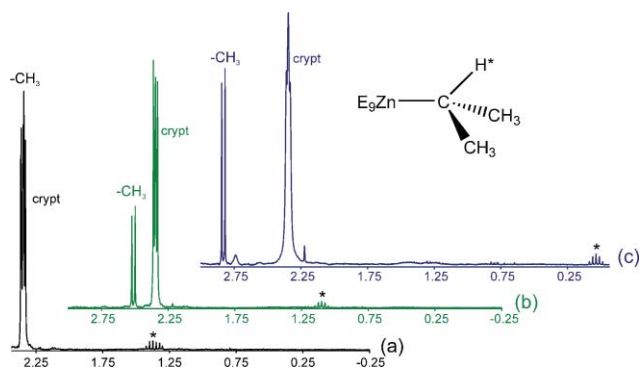


Fig. 4 Aliphatic regions of ^1H NMR spectra for (a) $[\text{Ge}_9\text{Zn}^i\text{Pr}]^{3-}$, (b) $[\text{Sn}_9\text{Zn}^i\text{Pr}]^{3-}$ and (c) $[\text{Pb}_9\text{Zn}^i\text{Pr}]^{3-}$ collected in d^5 -pyridine. Proton resonances corresponding to the methine protons have been highlighted with an asterisk. Data were recorded at 298 K at 300.17 MHz.

The Zn^iPr functionalised cluster anions **4–6** were also characterised in solution by electrospray mass-spectrometry. As noted earlier, the negative ion mode mass-spectra of the clusters re-

veal mass-envelopes corresponding to oxidised cluster anions, a common result of the ionization process. Extensive ionic pairing between the cluster anions and $[\text{K}(2,2,2\text{-crypt})]^+$ counteranions were also observed. The negative ion mode spectra of DMF solutions of **4–6** reveal mass envelopes corresponding to $[\text{E}_9\text{Zn}(\text{C}_6\text{H}_{11})]^-$ (Fig. 5; $m/z = 762.2$, 1175.7 and 1947.2 for **4**, **5** and **6**, respectively), peaks corresponding to the ion-paired species $\{[\text{K}(2,2,2\text{-crypt})][\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^-$ ($m/z = 1177.4$, 1175.7 and 1974.2 for **4**, **5** and **6**, respectively) and peaks corresponding to $\{[\text{K}(2,2,2\text{-crypt})]_2[\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^-$ ($m/z = 1592.5$ and 2006.8 for **4** and **5**, respectively). The positive ion mode spectra of solutions containing clusters **4** and **5** also reveals peaks corresponding to the $\{[\text{K}(2,2,2\text{-crypt})][\text{E}_9\text{Zn}(\text{C}_6\text{H}_5)]\}^+$ ion pairs at $m/z = 2428.0$ and 2840.0, respectively.

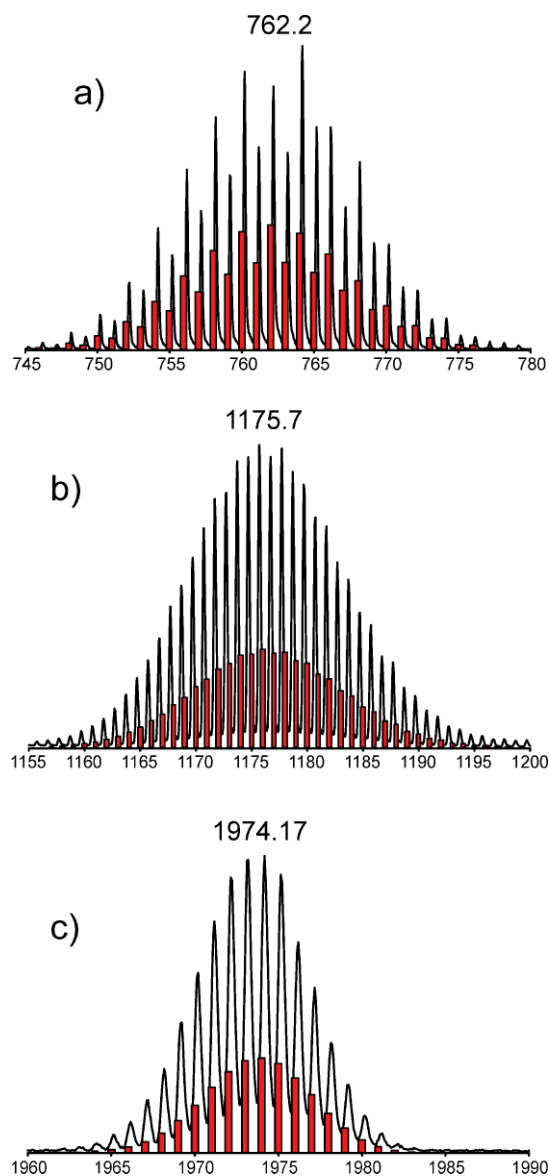


Fig. 5 Selected negative ion-mode electrospray mass-spectrometry peaks from DMF solutions containing **4–6**. Mass envelopes correspond to: (a) $[\text{Ge}_9\text{Zn}^i\text{Pr}]^-$, (b) $[\text{Sn}_9\text{Zn}^i\text{Pr}]^-$ and (c) $[\text{Pb}_9\text{Zn}^i\text{Pr}]^-$. Recorded experimental data are given in black with calculated isotopic distributions in red.

Conclusions

The six clusters reported in this manuscript provide further evidence of ability of deltahedral Zintl ions to reductively cleave weak metal–carbon bonds. Furthermore, the isolation of the first series of metal alkyl functionalized clusters highlights the versatility of this route towards novel mixed-metal clusters with *exo*-bonded organic substituents. Over the course of the last twenty years the coordination chemistry of deltahedral Zintl anions in solution has advanced enormously, however despite substantial progress the electronic influence of these clusters on coordinated metal atoms is still not well understood. The generation of libraries of novel functionalized Zintl ions and the elucidation of spectroscopic properties and chemical reactivity are crucial if we are to develop a clearer picture of the behaviour of such anions in solution. At present we are currently involved in a series of further studies probing the reactivity of post-transition metal functionalized Zintl ions towards a variety of small molecule substrates with the aim of gaining further insight into the chemical behaviour of such species in solution.

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